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Mediterranean intermittent rivers and ephemeral streams: Challenges in monitoring complexity

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Abstract

In most cases, intermittent rivers and ephemeral streams (IRES) form the principal surface water bodies in Mediterranean regions and people are heavily dependent on them. Strong reliance on such streams results in numerous pressures, and complex relationships are formed between their various natural and anthropogenic components, which creates numerous challenges to decision-makers who require a better understanding of IRES to establish effective measures to restore ecosystem functionality and implement changes, while maintaining human well-being. In this paper, we present a conceptual framework that identifies process cascades and links to identify the variables governing Mediterranean IRES and map their complex interrelations. Climate is the main external driver, and it affects IRES through precipitation, temperature, wind action, and geomorphic factors, which in turn influence the physical and biological attributes of IRES. Monitoring challenges and knowledge gaps are presented from the conceptual framework, highlighting the need to identify solutions for the monitoring and mapping the spatio-temporal variability of flow regimes and their influence on bio-physical variables and mapping and quantifying ecosystem services for integrated environmental management. Finally, we argue that remote sensing techniques, specifically, the combined use of unmanned aerial vehicles and structure from motion photogrammetry, are key to bridging gaps in understanding and meeting the challenges of managing Mediterranean IRES.

KEYWORDS

ecosystem services, environmental management, Mediterranean intermittent streams, remote sensing, structure-from-Motion photogrammetry, unmanned aerial vehicles

1 | INTRODUCTION

Temporal flows in intermittent rivers and ephemeral streams (hereafter, IRES; Datry, Bonada, & Boulton, 2017; Stubbington et al., 2018) produce complex dynamics between their abiotic and biotic environments. In the Mediterranean basin, IRES are the dominant surface water bodies (Bonada & Resh, 2013; Froebrich, Nikolaidis, Prat, & García-Roger, 2010), and people are heavily dependent upon them. Managers and planners developed numerous management measures

to maximize the exploitation of their services, such as the provisioning of water and sediment (Aristi et al., 2014; Bellin, van Wesemael, Meerkerk, Vanacker, & Barbera, 2009; Bonada & Resh, 2013; Cooper, Lake, Sabater, Melack, & Sabo, 2012; Moyle, 2014). However, without prior knowledge of the complex interrelationships between the natural (e.g., hydrological regime and habitats) and human variables (e.g., chemical contamination and introduction of alien species), implementing management interventions in IRES may undermine the delivery of other equally significant environmental services.

The ability of Mediterranean IRES (hereafter, referred to as Mediterranean IRES that are found in the Mediterranean basin) to provide numerous ecosystem services (ES) is often undervalued (Stubbington et al., 2018). These transient waterways are an important resource for agriculture through the provision of irrigation and rich soils. They serve as recharge points for underlying aquifers, and if properly maintained, they can serve as vital conduits for flood relief, especially in urbanized/populated areas. In some countries they are also valued for their provision of water and food such as through the provision of medicinal plants, herbs or provide grazing grounds for cattle (Steward, von Schiller, Tockner, Marshall, & Bunn, 2012). Mediterranean IRES are unique ecosystems that in some areas are biodiversity hotspots, harbouring numerous endemic species (Hershkovitz & Gasith, 2013). Flow intermittency encourages the development of complex habitats that support semi-aquatic and terrestrial species (Datry et al., 2017). When IRES dry out numerous ecological functions are influenced including processes related to organic matter and nutrient re-cycling (Datry, Boulton, et al., 2017). These dynamics are, however, still poorly researched (Datry, Boulton, et al., 2017).

Adequate planning and management of Mediterranean IRES necessitate a clear understanding of the links between the natural and human-derived variables operating at different temporal and spatial scales. Due to its spatial coverage, potential for high temporal repeat rates and ability to access previously unreachable locations, remote sensing has increasingly been utilized to map and monitor river environments (e.g., Carboneau, Fonstad, Marcus, & Dugdale, 2012; Marcus & Fonstad, 2010). Furthermore, novel remotely sensed techniques involving digital photogrammetry processing techniques such as structure from motion (SfM) combined with the use of unmanned aerial vehicles (UAVs) have opened up new opportunities for cost-effective, user-friendly remote sensing with high monitoring frequency and mapping of land surfaces changes on fine spatial scales (DeBell, Anderson, Brazier, King, & Jones, 2016; Hortobágyi, Corenblit, Steiger, & Piery, 2017).

Here we produce a conceptual framework (Figure 1) that tries to holistically capture the dynamism of Mediterranean IRES, highlighting the key process cascades, the various variables (categorized as *exogenous*, *physical* and *biological*), and their interactions within the system. We also identify the various challenges faced in monitoring and mapping IRES. Consequently, we argue that the use of remotely sensed applications such as UAVs and SfM photogrammetry is key to capture and monitor complexity in Mediterranean IRES. Consequently, the objectives of this paper are to (i) illustrate through a conceptual diagram the complex interactions in Mediterranean IRES, (ii) recognize key monitoring and mapping challenges in IRES, and (iii) advance the use of UAV and SfM photogrammetry to address these challenges.

2 | MONITORING CHALLENGES

Complexity in Mediterranean IRES is derived from the interrelationships between the multiple natural and human-induced abiotic and biotic variables. These relationships acting at multiple scales

(watercourse, subcatchment, and catchment) and the hierarchical influences on the functioning of Mediterranean IRES operating at distant spatial and temporal scales (Munné & Prat, 2004) create numerous challenges to environmental managers. Enhanced knowledge of the different variables and their interconnections is required to establish the measures that are needed to restore sustainable function and/or implement changes in the catchment area without altering current operation. In this section, we explore and present a conceptual framework (Figure 1) that adopts a comprehensive view of the variables and their interactions. The conceptual framework was established on a review of literature sources concerning the ecology, hydrology, ecohydrology, sociohydrology, hydrogeology, and environmental management of transient waterways in the Mediterranean basin. The framework helps us to explore the challenges in monitoring and mapping of Mediterranean IRES (section 2.2).

2.1 | Conceptual framework representing complexity and linked process cascades

Representing all the processes that are manifested in Mediterranean IRES through a holistic diagram is merely an unrealistic feat. The conceptual framework diagram (Figure 1) represented here is guided by the need to provide a simpler and more tangible visual of the (i) multiscale phenomena that are part of the Mediterranean IRES system represented by *exogenous variables* and the endogenous *physical* and *biological variables* and (ii) linkages amongst natural and/or human-derived variables influencing the system on a spatio-temporal scale. The linkages between the different variables are colour-coded, providing a broad explanation on the type of relationship being represented.

Exogenous variables, external to the catchment system, drive system changes directly influencing the endogenous *physical* and *biological variables*. Climate is the main external driver, and it varies across different latitudes and longitudes and is also influenced by humans through climate change. Climate affects IRES through four main process cascades driven by (i) *precipitation* patterns, types, frequency, and amounts; (ii) *temperature*; (iii) *wind* action (aeolian processes); and (iv) *geomorphic factors* that configure the landscape. The latter is included in the endogenous part of the diagram to reflect the shaping of landforms in different geographical locations and associated climatic zones. However, the geomorphic environment represented by *geology* and *fluvial geomorphology* in the *physical variables* section of the diagram.

is inherently influenced by the processes occurring at the catchment scale. All the "climatic" variables influence the management and planning frameworks adopted by decision-makers. In-turn, *planning and management* actions can, in one-way or another, directly and/or indirectly modify the physical and biological components of the system.

2.2 | The challenges

The complexity of Mediterranean IRES arises not only from the number and variability of the controlling variables but also from the

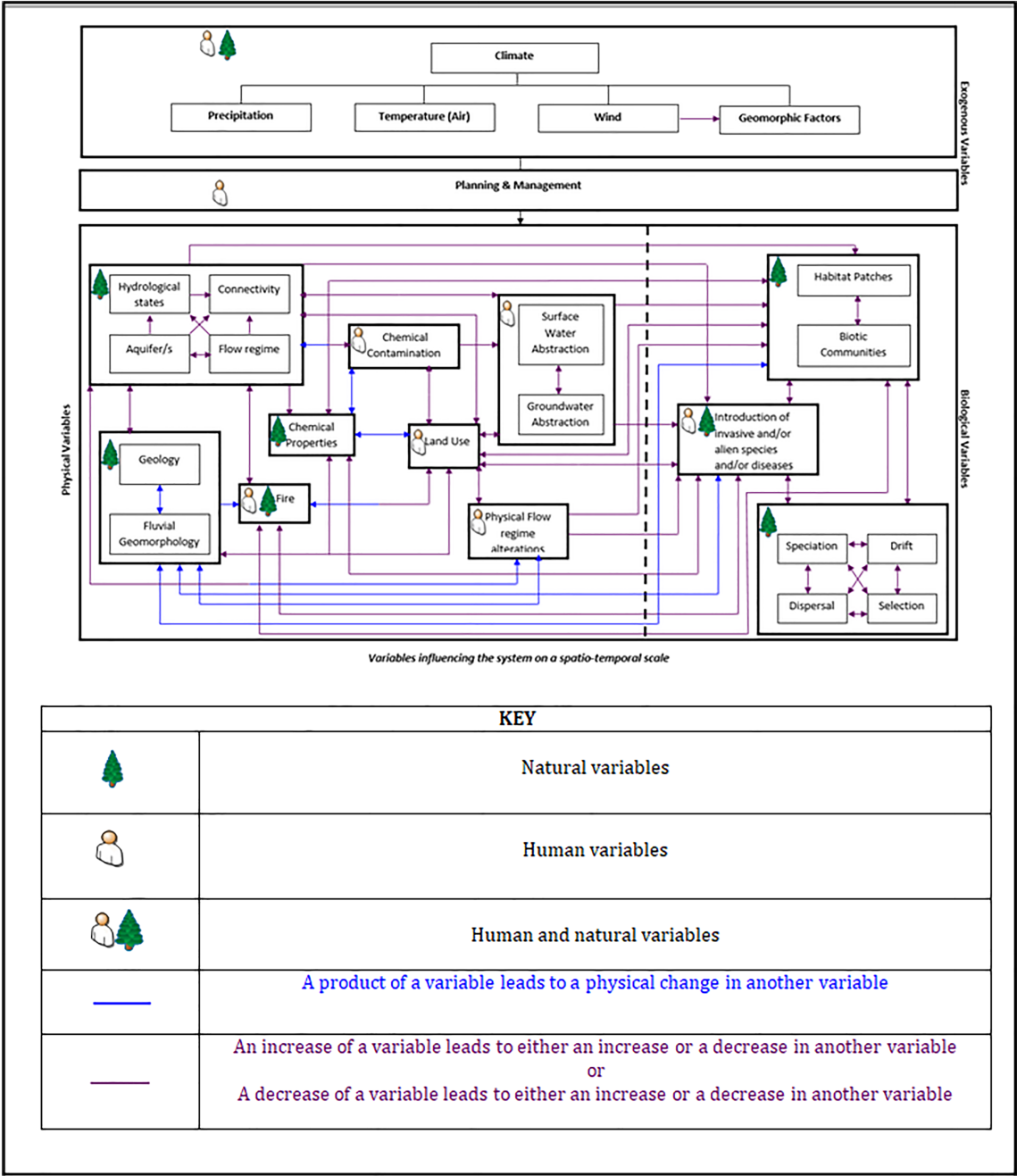


FIGURE 1 Conceptual framework compartmentalizing the natural and human-induced exogenous and indigenous variables that drive Mediterranean intermittent rivers and ephemeral streams. Endogenous variables are divided into physical and biological variables. Arrows depict the links between each compartment and variables within

interactions between them. Our conceptual model shows that climate is the main driver affecting all processes through precipitation, wind action, and temperature fluctuations, which are mediated by catchment geology. Each of these variables sets in motion a series of changes which alter the characteristics and behaviour of the other natural and anthropogenic elements in the catchment area. The conceptual framework sharpens the focus on the numerous monitoring and management challenges in Mediterranean IRES. In this section the challenges and existing knowledge gaps that we believe future research should focus on are presented.

Challenge 1: Monitoring and mapping the spatio-temporal variability of flow regimes and their influence on bio-physical variables

Various methods have been used to characterize the flow regime and other physical attributes of Mediterranean IRES. Gauging stations provide information about the frequency and duration of temporary flow using flow time series and facilitate a more specific definition of flow intermittency (González-Ferreras & Barquín, 2017; Snelder et al., 2013). However, most IRES catchments in the Mediterranean typically have few or no gauging stations. The installation of gauging

stations is not only costly but may be prohibitive due to active geomorphological processes and restricted accessibility (Beaufort, Carreau, & Sauquet, 2019; Beaufort, Lamouroux, Pella, Datry, & Sauquet, 2018). On the ground wet and dry mapping is sometimes used to compensate the absence of gauges (Datry, Fritz, & Leigh, 2016; Turner & Richter, 2011). This method is especially useful to qualitatively describe/capture the occurrence of the different hydrological states (Costigan et al., 2017). However, this procedure demands considerable time in the field and it is impractical when mapping is to be carried out in frequent intervals and over extensive areas.

With the advent of smartphones equipped with cameras and Global Positioning Systems, the monitoring of real-time hydrological states by appropriately trained citizen scientists, has become a promising method to advance the mapping of IRES (Costigan et al., 2017). Wet and dry mapping of the spatial extent of hydrological flows by citizen scientists can be carried out over large areas and at many different sites (Turner & Richter, 2011). Citizen scientists may also be trained to collect other important ecohydrological data such as the presence of different aquatic states, including pools or dry areas (Allen et al., 2019) that would otherwise be undetectable in gauged streams (Costigan, Jaegar, Goss, Fritz, & Goebel, 2016). Crowdwater, RIU.NET and Stream Tracker are examples of projects that crowdsource hydrological data from IRES through smartphone applications where users take geo-tagged photographs and use the app to digitally add staff gauges, eliminating the need for physical stream installations or manual measurements (Kampf et al., 2018; Prat et al., 2016). However, a prominent issue with surveying streams on foot is safety and lack of accessibility. To monitor the extent of perennial surface flow of the San Pedro River by citizen scientists, parts of the river were surveyed on horses or all-terrain vehicles (Turner & Richter, 2011). This, however, is not feasible in all IRES. In addition, network connections may be limited in low-income regions, hindering the accessibility and communication of data (Njue et al., 2019).

More research is required to capture and understand the variability of bio-physical processes at various spatial scales that are subject to natural and human pressures (Stella, Rodríguez-González, Dufour, & Bendix, 2012). Data capture and monitoring methodologies must be focused on providing solutions to quantify the dynamic responses of biota to the inherent hydrological transitions and other exogenous and indigenous environmental changes in Mediterranean IRES. Moreover, the geographical scope of both research and management should extend from a purely Mediterranean IRES approach, in a way that would address habitats from a biogeographical perspective, including corridor and catchment areas at various spatial scales. As advocated in Integrated River Basin Management (IRBM), a comprehensive approach of river management is needed to understand how the different flows in energy, matter, and ecosystems operate at various scales (Carbonneau et al., 2012; Fausch, Torgersen, Baxter, & Li, 2002). This would require the use of conceptual and practical tools that require the spatial quantification of a number of variables (such as dominating bio-physical conditions and ES) and the need to address how climate change will impact on the bio-physical conditions and

how scenarios can be constructed to look at potential impacts on Mediterranean IRES communities.

Challenge 2: Mapping and quantifying ecosystem services for integrated environmental management.

The over usage of aquifer and surface waters is both an economic and political burden for Mediterranean countries (Larned, Datry, Arscott, & Tockner, 2010; Nikolaidis et al., 2013). The functions provided by Mediterranean IRES continue to underline the importance of safeguarding and enhancing the services and goods provided by these systems. Considering that Mediterranean IRES play a part in peoples wellbeing, a balance needs to be struck between ensuring their proper management and meeting human needs. Arthington, Bernardo, and Ilhéu (2014) stress that instead of aiming to reach an unachievable original environmental state, management initiatives of temporary streams should be focused more on their capacity to provide numerous services to society. Mediterranean IRES have the potential to provide a variety of other supporting, regulating, provisioning and cultural services. However, these are mostly public goods that are not easily measured (e.g., nutrient recycling, biodiversity, sense of place, amongst others) and quantifiable in monetary terms (Gilvear, Spray, & Casas-Mulet, 2013). Historically, these services have in most cases been ignored when managing Mediterranean IRES (Boulton, 2014; Datry, Boulton, et al., 2017).

The availability of spatially explicit ES data at different resolutions and at catchment-wide scale is important for managers to prioritize interventions, identify problem areas and better understand synergies and trade-offs amongst the various ES in the system (Fu et al., 2015; Merenlender & Matella, 2013). However, most of these services are not always manifested and used within the same location or timeframes (Gilvear et al., 2013). A better understanding of *what*, *when*, and *where* are ecosystems services provided is cardinal to conserve and enhance them. Even though many ES in freshwater systems are dependent on flow (Willaarts, Volk, & Aguilera, 2012) a large number of environmental functions in IRES extend during the dry period, providing regulating and supporting services such as erosion regulation and alleviating flooding during storm events (Datry, Bonada, & Boulton, 2017; Koundouri, Boulton, Datry, & Souliotis, 2017). However, knowledge of the influence of alternating dry and wet conditions on the ecosystem processes that produce ES is still scarce (Datry, Boulton, et al., 2017; Koundouri et al., 2017). Altered flows may also be beneficial to local communities through the provision of additional/new services such as flood protection, water provisioning, and aquifer recharge to sustain rural and agricultural communities. Hence, in the absence of long-term flow data and with a desire to meet a particular set of ecological and social objectives, a recent historical baseline have to be used for the development of “designed” environmental flows that provide the needed ES (Acreman et al., 2014). This however necessitates a clear understanding of the relationships between flow variations and ecosystem functioning in the system.

The services provided and processes behind biogeochemical transitions have been discussed in the literature (Larned et al., 2010; Datry, Larned, & Tockner, 2014; Costigan et al., 2016; von Schiller et al., 2015; Arce et al., 2019). More research is needed on the benefits and activities of terrestrial biota during the dry periods and their interactions with aquatic species in terms of structural attributes such as soil characteristics (Arce et al., 2019), composition of vegetation and channel morphology and those primarily influencing functional attributes such as energy flow and nutrient cycling. Little is known on the impact of multiple stressors in temporary streams and their influence on ecosystem functionality (Navarro-Ortega et al., 2015). Even though Mediterranean IRES are naturally resilient to water shortage, they are particularly fragile to multiple stressors such as flow intermittency and anthropogenic pollution (Smeti et al., 2019). This is further complicated by evidence of pollutant accumulation during no flow periods (Karaouzas, Theodoropoulos, Vardakas, Kalogianni, & Skoulikidis, 2018) and multivariant stressor effects, stressor interactions, and stressor impacts to the different environment variables along the river's continuum (Segurado, Almeida, Neves, Ferreira, & Branco, 2018). Furthermore, environmental managers have to deal with a constantly changing anthropogenic environment with the introduction/discovery of new pollutants. For example in Malta, 82% of samples collected from ephemeral pools in undeveloped areas between 2011 and 2013 contained levels of perchlorate which are well above the "natural" values, resulting from the only known strong source of percholate, the burning of fireworks (Pace & Vella, 2019). The impact of this contamination to the biogeochemical characteristics of IRES is still unknown.

3 | ADVANCING THE USE OF UAV AND SFM PHOTOGRAMMETRY TO IMPROVE MONITORING OF MEDITERRANEAN IRES

The challenges presented share a common theme: the difficulty in monitoring and mapping of the spatio-temporal variability of water flows is hindering the better understanding of its impact on the biophysical variables, ES and ecological resilience of Mediterranean IRES. Water is the most essential component for ecosystem functioning (Sponseller, Heffernan, & Fisher, 2013; Willaarts et al., 2012) and in IRES the delivery of ES is most likely to be greatly influenced by the duration, timing, frequency and extent of the different hydrological states (Datry, Boulton, et al., 2017). Understanding this synergy is crucial especially in the Mediterranean where IRES are exposed to multiple anthropogenic stressors altering their flow regime (Skoulikidis et al., 2016). This will be further exacerbated by climate change where long-term reduction in the number of flow days and decrease in stream flow trends have already been recorded in the Mediterranean region (De Girolamo, Bouraoui, Buffani, Pappagallo, & Lo Porto, 2017; Erol & Randhir, 2012; Garcia, Amengual, Homar, & Zamora, 2017; Giorgi & Lionello, 2008; Moran-Tejeda, Lopez-Moreno, Ceballos-Barbancho, & Vicente-Serrano, 2011; Nerantzaki et al., 2015; Schneider, Laizé, Acreman, & Flörke, 2013). Thus, an enhanced

capability to monitor the spatio-temporal variation of the hydrological states in relation to other physical parameters is imperative for this understanding.

Remote sensing offers an efficient approach to monitor large areas with spatially continuous coverage of fluvial systems (Tamminga, Hugenholtz, Eaton, & Lapointe, 2015; Woodget, Carbonneau, Visser, & Maddock, 2015), which cannot be achieved by traditional means. This is especially applicable in Mediterranean IRES where the transitioning between arid and wet periods form different ecohydrological conditions along the stream continuum. Advances in remote sensing platforms, sensor technology, and data processing/analysis techniques now means that a range of fluvial metrics (e.g., channel morphology/flow state; Dietrich, 2016; Woodget, Visser, Maddock, & Carbonneau, 2016; grain size; Carbonneau, Bergeron, & Lane, 2005; Black, Carbonneau, Church, & Warburton, 2014; sediment load; Long & Pavelsky, 2013; Olmanson, Brezonik, & Bauer, 2013; depth/bathymetry, Legleiter, 2013; McKean et al., 2009; water temperature; Torgersen, Faux, McIntosh, Poage, & Norton, 2001; Dugdale, 2016; and aquatic/riparian vegetation characteristics; Bertoldi, Gurnell, & Drake, 2011; Flynn & Chapra, 2014) can now be assembled with relative ease. Although conventional satellite imagery and aerial photography can be costly, freely available high spatial resolution aerial photographs (Google Earth and other virtual globes) may be used to characterize hydrologic states (e.g., Legleiter, 2013) and/or map ES (Large & Gilvear, 2015). However, information is only limited to specific capture dates (Gallart et al., 2012., Gallart et al., 2016, Costigan et al., 2017) and is therefore not well-suited to site-specific studies at small scales (Tamminga et al., 2015). This is especially true in the Mediterranean, where most IRES are usually not more than a few metres wide. Use of these "conventional" remote sensing techniques has, therefore, found limited use in regions dominated by Mediterranean IRES.

Recently however, novel remotely sensed - digital photogrammetry processing techniques such as SfM, have enabled scientists to work directly with spatially explicit continuums of river data (Woodget, Austrums, Maddock, & Habit, 2017). SfM photogrammetry can be applied at multiple temporal and spatial scales and at extremely detailed resolutions, allowing for novel opportunities for the understanding of earth surface processes such as 4D (three spatial dimensions and one temporal dimension) reconstruction of landscape dynamics (Eltner et al., 2016). SfM photogrammetry develops 3D structures from 2D image sequences by automatically matching conjugate points between scenes acquired from different angles (Fonstad, Dietrich, Courville, Jensen, & Carbonneau, 2013; Woodget et al., 2015). SfM is increasingly becoming more popular amongst scientists due to its affordability and the existence of easy-to-use commercial and open source software applications that allow even nonscientists to build three-dimensional models of surface features which are subjects of scientific research (Wróżyński, Pyszny, Sojka, Przybła, & Murat-Blązejewska, 2017). SfM has been used in a vast array of ecohydrological and river management applications such as for the observation of different river stages (Duró, Crosato, Kleinhans, & Uijttewaalt, 2018; Niedzielski, Witek, & Spallek, 2016), analysis of

feedbacks between fluvial geomorphology and riparian vegetation (Hortobágyi et al., 2017); floodplain inundation mapping (Schumann, Muhlhausen, & Andreadis, 2019) and river management and restoration (Kubota, Kawai, & Kadotani, 2017; Marteau, Vericat, Gibbins, Batalla, & Green, 2017).

The development of SfM technology has been accompanied by the emergence of UAVs, which provided scientists with cost-effective, and user-friendly remote sensing with high monitoring frequency and mapping at fine spatial scales (Hortobágyi et al., 2017). Figure 2 provides an example of 3 cm resolution orthomosaic produced from a consumer-grade quadcopter drone and SfM photogrammetry application of a Mediterranean ephemeral stream. To the authors' knowledge, the combined use of SfM and UAVs to investigate the spatio-temporal variability of alternating dry and wet stages in Mediterranean IRES has

not yet been adopted. However, the combination of these techniques is particularly suited to Mediterranean IRES, where the ability to respond quickly to wetting and drying events is of paramount importance. Contrary to other remotely sensed applications, UAVs are not limited to specific capture dates. UAV flight missions can be carried out much more frequently, providing added solutions for the understanding of complex processes in IRES. For example, the generation of 4D models depicting the evolution of the different hydrological states on successive dates at different spatial scales and with appropriate resolutions for ecohydrological monitoring would provide scientists with an unparalleled perspective of the spatio-temporal dynamics of the different hydrologic states in Mediterranean IRES.

Combined with light-weight instruments that cover the visible to thermal spectrum including multispectral or hyperspectral sensors,

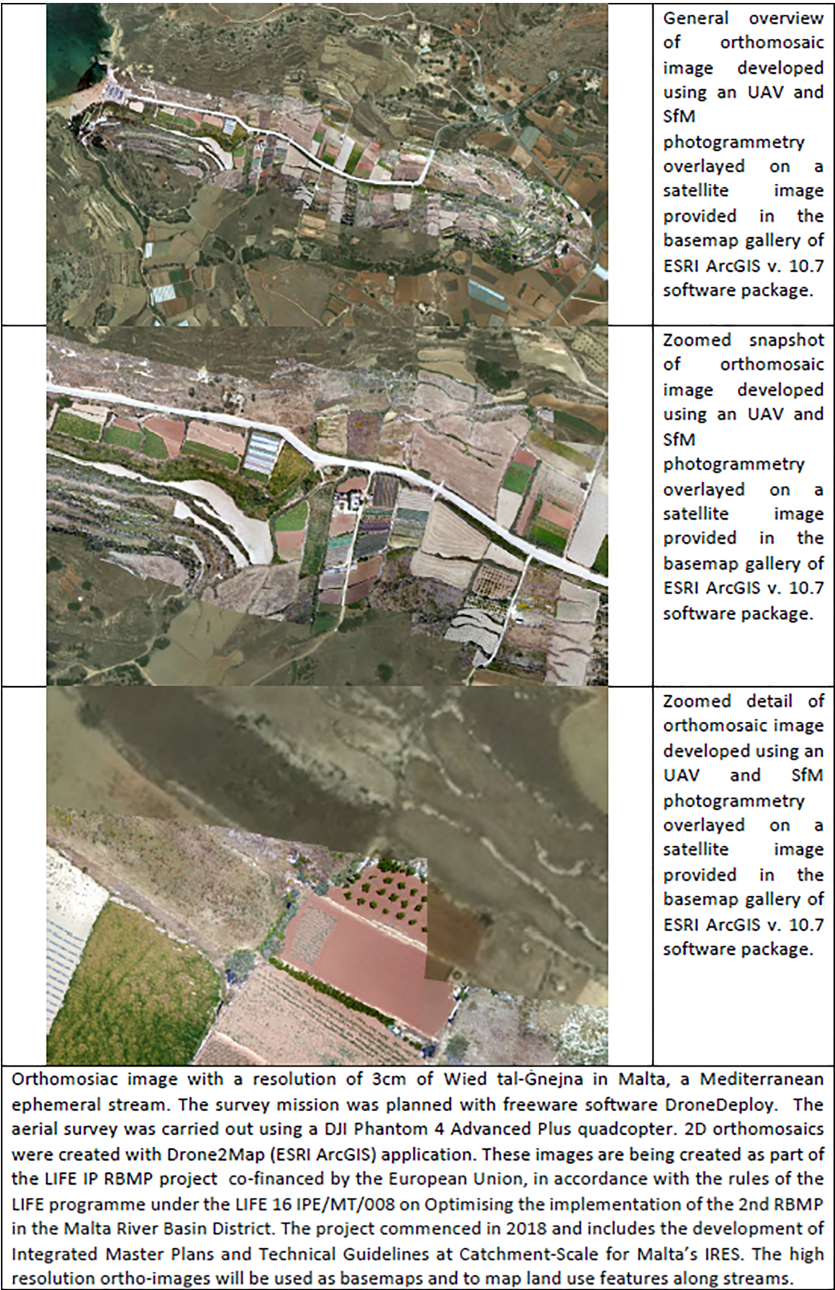


FIGURE 2 High-resolution images of a Mediterranean ephemeral stream developed with a commercial unmanned aerial vehicle (UAV) and structure from motion (SfM) photogrammetry software compared with satellite imagery provided by ESRI that includes 15-m TerraColor imagery at small and midscales and 2.5-m SPOT imagery for the world

miniature RADAR, passive microwave radiometers, and light detection and ranging (Evaraerts, 2008), UAVs also have potential for monitoring other variables occurring along Mediterranean IRES. Although still in their infancy (in terms of their mounting on UAVs), these technologies have the potential to provide scientists with key tools to further comprehend the influence of hydrological intermittence on other stream biophysical variables at greater resolutions and more frequent intervals (Politi, Rowan, & Cutler, 2016). For example, aquatic and terrestrial habitat changes and sediment transfer in Mediterranean IRES, which are otherwise difficult to detect from the use of "conventional" remote sensing platforms, can be monitored. Many of the river remote sensing methodologies that were previously only available to conventional remote sensing platforms can now be achieved using UAV-derived data. For example, recent studies demonstrate the maturity of UAV-mounted multispectral sensors for mapping riparian vegetation in arid regions (e.g., McCabe et al., 2017), a technique that was previously limited to airborne or satellite approaches. Although there are currently no further peer-reviewed examples of the use of UAV-based multispectral sensors for the extraction of river habitat data, the authors are aware of efforts to quantify river water quality parameters (e.g., suspended sediment, algal concentrations) from such instruments. Similarly, although no published studies of UAV-based river temperature monitoring currently exist, several examples using thermal infrared imaging are present in the grey literature (e.g., Jensen, Neilson, McKee, & YangQuan, 2012). The ability to acquire multispectral or thermal infrared imaging data from UAVs represents a key boon for researchers studying Mediterranean IRES where accurate data on water quality parameters (e.g., sediment yield, temperature) would fill two key data requirements.

UAVs are also opening possibilities for the development of new methods and techniques not previously applicable to conventional remote sensing, which could also have benefits to the study and integrated management of Mediterranean IRES. For instance, surface flow velocity measurement has long been difficult to achieve using conventional remote sensing as velocity-sensing algorithms (e.g., particle image velocimetry or PIV; Daigle, Bérubé, Bergeron, & Matte, 2013) require multitemporal datasets. However, the ability of drones to capture image sequences while hovering or flying slowly means that velocity mapping in river environments is now a reality (e.g., Bolognesi et al., 2017; Perks, Russell, & Large, 2016). Similarly, although initial experiments with UAVs necessitated lower resolution imaging sensors and was often associated with decreased image quality (due to motion blur; Sieberth, Wackrow, & Chandler, 2014), the increasing miniaturization of camera technology coupled with enhanced gyro-stabilized mounts (gimbals; Gasparovic & Jurjevic, 2017) means that in many cases, UAVs are able to provide data at even higher resolution than conventional airborne platforms. Use of SfM and UAV is also permitting the increase in frequency of high-resolution topographic surveys and to monitor geomorphic changes, especially in inaccessible areas such as mountain catchments (Cucchiario et al., 2018). These advances are both well suited for Mediterranean IRES environments, where the increased spatial resolution of modern UAV sensors coupled to the ability to capture data either on an ad-hoc basis or at high temporal

frequency, and to reach restricted areas, means that, even in very small Mediterranean IRES channels, observations can be obtained at an appropriate scale of reference.

The current revolution in UAV technology has substantial implications for the improved monitoring and management of Mediterranean IRES. Nonetheless, there are a few disadvantages to the use of such technology. Flights are dependent on weather conditions where optimal operational wind speeds for most aircrafts are below a threshold of 24–32 km per hour and atmospheric moisture may damage the UAV if not adequately waterproofed (DeBell et al., 2016). As with all remote sensing-based approaches, issues of data calibration/validation remain a key impediment to their uptake. Most forms of remote sensing data still require some form of ground validation/calibration, and this is currently no less true for UAV/SfM approaches. Given the rapidity with which conditions change in Mediterranean IRES environments, a key challenge will be the collection of ground validation/calibration data within the same spatio-temporal window as UAV surveys are conducted; without such data, it will be very difficult to assess the validity of UAV-derived measurements in Mediterranean IRES. Similarly, current error-assessment criteria commonly used by scientists can be difficult to apply to the "riverscape-style" continuums of data acquired using UAVs, where data quality (accuracy) is often a trade-off for greatly increased data quantity. For example, when mapping densely vegetated areas, UAV-SfM produces point cloud data based only on the canopy surface that is visible from sky, unlike light detection and ranging data, which penetrate vegetation and tree cover and provide details of the terrain (Jayathunga, Owari, & Tsuyuki, 2018). Researchers working with such measurements (such as calculating canopy height) should therefore look to improve error-assessment approaches, making them easier to apply to these large spatio-temporal datasets. Another key concern regarding the application of UAVs in Mediterranean IRES is one of strict legislation regarding UAV flights over populated areas (Schumann et al., 2019). Because Mediterranean IRES are often highly modified environments lying close to human settlement, the use of UAV approaches may be limited either by pragmatic health and safety concerns or by legal impediments to flight over population centres. It will therefore be important to seek methods to fill/interpolate data gaps induced by the inability to acquire data in urban areas, possibly through the fusion of UAV data with ground measurements and observations from "conventional" remote sensing platforms. Nonetheless, this, alongside other potential concerns should not detract from the obvious benefits that UAVs-based remote sensing (and associated techniques like SfM) bring to meeting the challenge of improved mapping and monitoring of Mediterranean IRES environments.

4 | CONCLUSION

Current methods used for the monitoring and mapping of the spatio-temporal variability of flows are limiting in identifying links with the bio-physical variables and provision of ES in IRES. This is further complicated by the impact of multiple human-induced stressors in

Mediterranean IRES and their influence on the streams' functions and processes. With their unparalleled ability to deliver, at affordable costs, fine spatial and temporal resolution data, whilst reaching areas which are otherwise inaccessible, SfM photogrammetry applications combined with UAV-based technologies are an ideal solution to better understand complexity in Mediterranean IRES. There are a number of challenges that still need to be addressed including the advancement of weatherproof UAVs, production of accurate remotely-sensed data without the need for the collection of ground-control points to validate measurements, improving error-assessment approaches in large-scale spatio-temporal datasets, and development of legal frameworks ensuring safe UAV use for both the general public and the users whilst not inhibiting scientific research. However, this paper has explained that there is great potential for exploring UAV and SfM use to monitor, better understand and manage Mediterranean IRES.

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